CENTRAL BOARD OF SECONDARY EDUCATION

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COMPUTER SCIENCE (083)

PROJECT FILE

On

HOSPITAL MANAGEMENT SYSTEM USING PYTHON AND MYSQL

Session: 2026-27

Submitted By:
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ACKNOWLEDGEMENT

I wish to express my deep sense of gratitude and indebtedness to our learned teacher Pankaj Bhilingwal, PGT Computer Science, J.N.V Tarikhet Almora for his invaluable help, advice and guidance in the preparation of this project file.

I am also greatly indebted to our principal D.S Rawat and school authorities for providing me with the facilities and requisite laboratory conditions for making this project file.

I also extend my thanks to a number of teachers, my classmates and friends who helped me to complete this project file successfully.

Signature of Student

CERTIFICATE

This is to certify that,

S. No	Name	CBSE Roll

Student of Class XII, J.N.V Tarikhet Almora has completed the PROJECT FILE titled “Hospital Management System Using Python and MySQL” during the academic year 2026-27 towards partial fulfilment of credit for the Computer Science Project evaluation of CBSE and submitted satisfactory report, as compiled in the following pages, under my supervision.

Internal Examiner

Signature

External Examiner

Signature

PREFACE

In the field of Computer Science, the term PROJECT refers to a computerization work that is completely performed by using a computer. Today almost every field is computerized, and the different tasks of every field are done using computers and appropriate computer programs. Such programs and software are developed in the form of Projects.

Learning Python in schools is a part of our syllabus, and according to the syllabus worked out by the CBSE, to which my school is affiliated, I enjoyed doing project work for the session 2026-27. I learned a lot, especially learning by doing was a great experience. The report is a summary of what I observed and learnt in school. My humble thanks go to all those concerned with my project.

I also feel that it is the right time to thank my teacher, who guided me during my project, and my parents, friends and project partners for the constant assistance and cooperation extended by them from time to time.

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About Project

Hospital Management System Using Python and MySQL

The Hospital Management System is a menu-driven program developed in Python with MySQL as the back-end database. It helps the reception desk of a hospital to keep the records of doctors and patients and to manage appointments.

The data is stored in three related tables - doctors, patients and appointments. The program connects Python with MySQL using the mysql.connector module and performs all common SQL operations: creating tables, inserting, searching, updating, joining tables and generating reports with aggregate functions.

Project Overview

Objective:

The objective of this project is to learn how a Python program works with a relational database. It replaces paper registers at a hospital reception with a computerized system that is fast, accurate and gives useful reports like the number of appointments per department and fees collected by each doctor.

Key Features:

1. **Add / Show Doctors:** Add a new doctor and view doctors of all departments or of one department.
2. **Register Patient:** Store patient ID, name, age, gender and phone number.
3. **Search Patient:** Search patients by full or partial name using the LIKE operator.
4. **Book Appointment:** Book an appointment of a patient with a doctor on a date.
5. **Cancel Appointment:** Change the status of an appointment to 'Cancelled' using UPDATE.
6. **Show All Appointments:** Display appointments with patient name, doctor name and department by joining three tables.
7. **Reports:** Appointments per department, fees collected by each doctor, and patient age statistics using GROUP BY, COUNT, SUM, MIN, MAX and AVG.
8. **Error Handling:** Shows a proper message for wrong input or database errors (e.g. duplicate patient ID).

Technology Stack

Programming Language: Python 3

Database: MySQL

Modules Used: mysql.connector

mysql.connector.connect() creates the connection, cursor() gives a cursor object, execute() runs SQL queries, fetchone() and fetchall() read the results, commit() saves changes and rowcount tells how many rows were changed. All values are passed to queries with %s placeholders, which is the safe and correct way of sending user input to MySQL.

Files in this Project

File	Purpose
hospital.py	Main program (menu, database connection and queries)

hospital_sample_data.sql	SQL script to create the database with sample records
flowchart.py	Python program that draws the flowchart using graphviz
flowchart.png	Flowchart image
img/	Screenshots of the output

How to Run the Program

1. Install Python 3 and MySQL Server.
2. Install the connector module: pip install mysql-connector-python
3. Open hospital.py and change PASSWORD = "admin" to your MySQL root password.
4. Run hospital.py. The database, tables and five doctors are created automatically on the first run.
5. (Optional) To load sample patients and appointments, run in MySQL: SOURCE hospital_sample_data.sql;

Syllabus Concepts Used

This project covers the following topics of the CBSE Class XII Computer Science (083) syllabus:

Concept	Where it is used in the project
Python-MySQL connectivity	connect(), cursor(), execute(), executemany(), commit(), close()
Fetching results	fetchone() for COUNT(*), fetchall() for lists of records, rowcount, lastrowid
DDL commands	CREATE DATABASE, CREATE TABLE with PRIMARY KEY, NOT NULL, DEFAULT, AUTO_INCREMENT, FOREIGN KEY
DML commands	INSERT (add doctor/patient/appointment), UPDATE (cancel appointment)
SELECT with clauses	WHERE, LIKE, ORDER BY, GROUP BY
Aggregate functions	COUNT(), SUM(), MIN(), MAX(), AVG() in reports()
Joins	Equi-join of appointments, patients and doctors tables
Parameterized queries	Values passed with %s placeholders, e.g. WHERE name LIKE %s
Exception handling	try-except for ValueError and mysql.connector.Error

Database Design

Database name: hospital. The program creates the database and the following three tables automatically.

Table: doctors

Stores details of doctors.

Column	Data Type	Constraint / Description
doc_id	INT	PRIMARY KEY
name	VARCHAR(40)	NOT NULL
department	VARCHAR(30)	Department of the doctor
fees	INT	Consultation fees

Table: patients

Stores details of registered patients.

Column	Data Type	Constraint / Description
pat_id	INT	PRIMARY KEY
name	VARCHAR(40)	NOT NULL
age	INT	Age of patient
gender	CHAR(1)	M / F
phone	VARCHAR(10)	Mobile number

Table: appointments

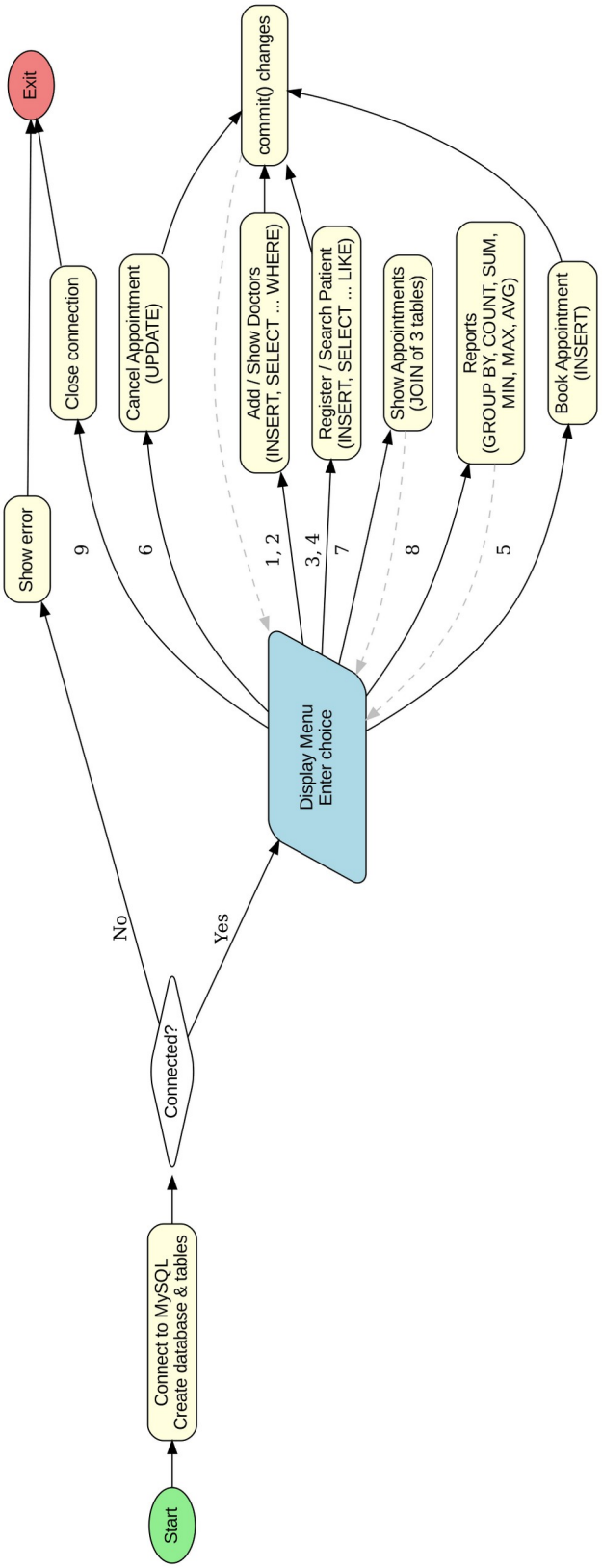
Stores which patient meets which doctor on which date.

Column	Data Type	Constraint / Description
app_id	INT	PRIMARY KEY, AUTO_INCREMENT
pat_id	INT	FOREIGN KEY references patients(pat_id)
doc_id	INT	FOREIGN KEY references doctors(doc_id)
app_date	DATE	Date of appointment
status	VARCHAR(10)	Booked / Cancelled (DEFAULT 'Booked')

Relationship between Tables

The appointments table connects the other two tables. Its pat_id column is a foreign key referring to patients and its doc_id column is a foreign key referring to doctors. So one patient can have many appointments and one doctor can have many appointments. To show patient and doctor names together, the program joins the three tables using the condition a.pat_id = p.pat_id AND a.doc_id = d.doc_id.

Flow Chart


Start
Connect to MySQL
Create database & tables
Connected?
No
Yes
Show error
Exit
Display Menu
Enter choice
9
6
1, 2
3, 4
7
8
5
Close connection
Cancel Appointment
(UPDATE)
Add / Show Doctors
(INSERT, SELECT ... WHERE)
Register / Search Patient
(INSERT, SELECT ... LIKE)
Show Appointments
(JOIN of 3 tables)
Reports
(GROUP BY, COUNT, SUM,
MIN, MAX, AVG)
Book Appointment
(INSERT)
commit() changes

Project Code

```
# Hospital Management System
# Python + MySQL interface (mysql.connector)
# Tables: doctors, patients, appointments

import mysql.connector

HOST = "localhost"
USER = "root"
PASSWORD = "admin"      # change to your MySQL password
DATABASE = "hospital"


def connect_database():
    # Create the database (first time) and connect to it
    con = mysql.connector.connect(host=HOST, user=USER, password=PASSWORD)
    cur = con.cursor()
    cur.execute("CREATE DATABASE IF NOT EXISTS " + DATABASE)
    con.close()
    con = mysql.connector.connect(host=HOST, user=USER,
                                  password=PASSWORD, database=DATABASE)
    return con


def create_tables(con):
    cur = con.cursor()
    cur.execute("""CREATE TABLE IF NOT EXISTS doctors (
                   doc_id INT PRIMARY KEY,
                   name VARCHAR(40) NOT NULL,
                   department VARCHAR(30),
                   fees INT)""")
    cur.execute("""CREATE TABLE IF NOT EXISTS patients (
                   pat_id INT PRIMARY KEY,
                   name VARCHAR(40) NOT NULL,
                   age INT,
                   gender CHAR(1),
                   phone VARCHAR(10))""")
    cur.execute("""CREATE TABLE IF NOT EXISTS appointments (
                   app_id INT PRIMARY KEY AUTO_INCREMENT,
                   pat_id INT,
                   doc_id INT,
                   app_date DATE,
                   status VARCHAR(10) DEFAULT 'Booked',
                   FOREIGN KEY (pat_id) REFERENCES patients(pat_id),
                   FOREIGN KEY (doc_id) REFERENCES doctors(doc_id))""")

    # Add some doctors the first time so the program can be used at once
    cur.execute("SELECT COUNT(*) FROM doctors")
    if cur.fetchone()[0] == 0:
        doctors = [(101, "Dr. R.K. Sharma", "Cardiology", 800),
                   (102, "Dr. Meena Joshi", "Pediatrics", 500),
                   (103, "Dr. Anil Rawat", "Orthopedics", 600),
                   (104, "Dr. Sunita Bisht", "General", 300),
                   (105, "Dr. Vikas Pant", "Cardiology", 700)]
        cur.executemany("INSERT INTO doctors VALUES (%s, %s, %s, %s)", doctors)
        con.commit()


# ---------------- Doctors ----------------

def add_doctor(con):
    cur = con.cursor()
    doc_id = int(input("Enter Doctor ID: "))
    name = input("Enter Doctor Name: ")
    dept = input("Enter Department: ")
    fees = int(input("Enter Consultation Fees: "))
    cur.execute("INSERT INTO doctors VALUES (%s, %s, %s, %s)",
                (doc_id, name, dept, fees))
    con.commit()
```

```
    print("Doctor added successfully!")


def show_doctors(con):
    cur = con.cursor()
    dept = input("Enter Department (press Enter for all): ")
    if dept == "":
        cur.execute("SELECT * FROM doctors ORDER BY department")
    else:
        cur.execute("SELECT * FROM doctors WHERE department = %s", (dept,))
    rows = cur.fetchall()
    if len(rows) == 0:
        print("No doctors found.")
        return
    print("%-6s %-20s %-14s %6s" % ("ID", "Name", "Department", "Fees"))
    print("-" * 50)
    for r in rows:
        print("%-6d %-20s %-14s %6d" % r)


# ---------------- Patients ----------------

def add_patient(con):
    cur = con.cursor()
    pat_id = int(input("Enter Patient ID: "))
    name = input("Enter Patient Name: ")
    age = int(input("Enter Age: "))
    gender = input("Enter Gender (M/F): ").upper()
    phone = input("Enter Phone Number: ")
    cur.execute("INSERT INTO patients VALUES (%s, %s, %s, %s, %s)",
                (pat_id, name, age, gender, phone))
    con.commit()
    print("Patient registered successfully!")


def search_patient(con):
    cur = con.cursor()
    name = input("Enter name (or part of name) to search: ")
    cur.execute("SELECT * FROM patients WHERE name LIKE %s", ("%" + name + "%",))
    rows = cur.fetchall()
    if len(rows) == 0:
        print("No patient found.")
        return
    print("%-6s %-20s %4s %6s %-10s" % ("ID", "Name", "Age", "Gender", "Phone"))
    print("-" * 50)
    for r in rows:
        print("%-6d %-20s %4d %6s %-10s" % r)


# ---------------- Appointments ----------------

def book_appointment(con):
    cur = con.cursor()
    pat_id = int(input("Enter Patient ID: "))
    doc_id = int(input("Enter Doctor ID: "))
    app_date = input("Enter Date (YYYY-MM-DD): ")
    cur.execute("INSERT INTO appointments (pat_id, doc_id, app_date) VALUES (%s, %s,
%s)",
                (pat_id, doc_id, app_date))
    con.commit()
    print("Appointment booked! Appointment ID:", cur.lastrowid)


def cancel_appointment(con):
    cur = con.cursor()
    app_id = int(input("Enter Appointment ID to cancel: "))
    cur.execute("UPDATE appointments SET status = 'Cancelled' WHERE app_id = %s",
(app_id,))
    con.commit()
    if cur.rowcount > 0:
        print("Appointment", app_id, "cancelled.")
```

```
    else:
        print("No appointment found with ID", app_id)


def show_appointments(con):
    cur = con.cursor()
    # JOIN of three tables
    cur.execute("""SELECT a.app_id, p.name, d.name, d.department, a.app_date,
a.status
                   FROM appointments a, patients p, doctors d
                   WHERE a.pat_id = p.pat_id AND a.doc_id = d.doc_id
                   ORDER BY a.app_date""")
    rows = cur.fetchall()
    if len(rows) == 0:
        print("No appointments found.")
        return
    print("%-4s %-16s %-18s %-12s %-11s %-9s" %
          ("ID", "Patient", "Doctor", "Department", "Date", "Status"))
    print("-" * 75)
    for r in rows:
        print("%-4d %-16s %-18s %-12s %-11s %-9s" %
              (r[0], r[1], r[2], r[3], str(r[4]), r[5]))


# ---------------- Reports ----------------

def reports(con):
    cur = con.cursor()
    print("\n--- Appointments per Department (GROUP BY) ---")
    cur.execute("""SELECT d.department, COUNT(*)
                   FROM appointments a, doctors d
                   WHERE a.doc_id = d.doc_id AND a.status = 'Booked'
                   GROUP BY d.department""")
    for r in cur.fetchall():
        print("%-14s : %d" % r)

    print("\n--- Fees Collected by each Doctor (SUM) ---")
    cur.execute("""SELECT d.name, SUM(d.fees)
                   FROM appointments a, doctors d
                   WHERE a.doc_id = d.doc_id AND a.status = 'Booked'
                   GROUP BY d.name
                   ORDER BY SUM(d.fees) DESC""")
    for r in cur.fetchall():
        print("%-20s : Rs. %d" % (r[0], r[1]))

    print("\n--- Patient Statistics (MIN, MAX, AVG) ---")
    cur.execute("SELECT COUNT(*), MIN(age), MAX(age), AVG(age) FROM patients")
    r = cur.fetchone()
    print("Total Patients:", r[0])
    if r[0] > 0:
        print("Youngest:", r[1], "| Oldest:", r[2], "| Average Age:",
round(float(r[3]), 1))


def main():
    try:
        con = connect_database()
    except mysql.connector.Error as e:
        print("Could not connect to MySQL:", e)
        return
    create_tables(con)

    while True:
        print("\n======= HOSPITAL MANAGEMENT SYSTEM =======")
        print("1. Add Doctor")
        print("2. Show Doctors")
        print("3. Register Patient")
        print("4. Search Patient")
        print("5. Book Appointment")
        print("6. Cancel Appointment")
        print("7. Show All Appointments")
```

```
        print("8. Reports")
        print("9. Exit")
        choice = input("Enter your choice: ")
        try:
            if choice == "1":
                add_doctor(con)
            elif choice == "2":
                show_doctors(con)
            elif choice == "3":
                add_patient(con)
            elif choice == "4":
                search_patient(con)
            elif choice == "5":
                book_appointment(con)
            elif choice == "6":
                cancel_appointment(con)
            elif choice == "7":
                show_appointments(con)
            elif choice == "8":
                reports(con)
            elif choice == "9":
                con.close()
                print("Database connection closed. Goodbye!")
                break
            else:
                print("Invalid choice! Please try again.")
        except ValueError:
            print("ERROR: Please enter numbers where required.")
        except mysql.connector.Error as e:
            print("Database ERROR:", e.msg)


main()
```

Output

Output 1: Doctors List

Python Shell - Doctors List

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 2
Enter Department (press Enter for all):
ID     Name                 Department       Fees
-------------------------------------------------
101    Dr. R.K. Sharma      Cardiology        800
105    Dr. Vikas Pant       Cardiology        700
104    Dr. Sunita Bisht     General           300
103    Dr. Anil Rawat       Orthopedics       600
102    Dr. Meena Joshi      Pediatrics        500

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 2
Enter Department (press Enter for all): Cardiology
ID     Name                 Department       Fees
-------------------------------------------------
101    Dr. R.K. Sharma      Cardiology        800
105    Dr. Vikas Pant       Cardiology        700
```

Output 2: Registering Patients

Python Shell - Registering Patients

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 3
Enter Patient ID: 1
Enter Patient Name: Ramesh Negi
Enter Age: 45
Enter Gender (M/F): M
Enter Phone Number: 9876543210
Patient registered successfully!

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 3
Enter Patient ID: 2
Enter Patient Name: Kavya Bisht
Enter Age: 8
Enter Gender (M/F): F
Enter Phone Number: 9412345678
Patient registered successfully!
```

Output 3: More Patients and Duplicate ID Error

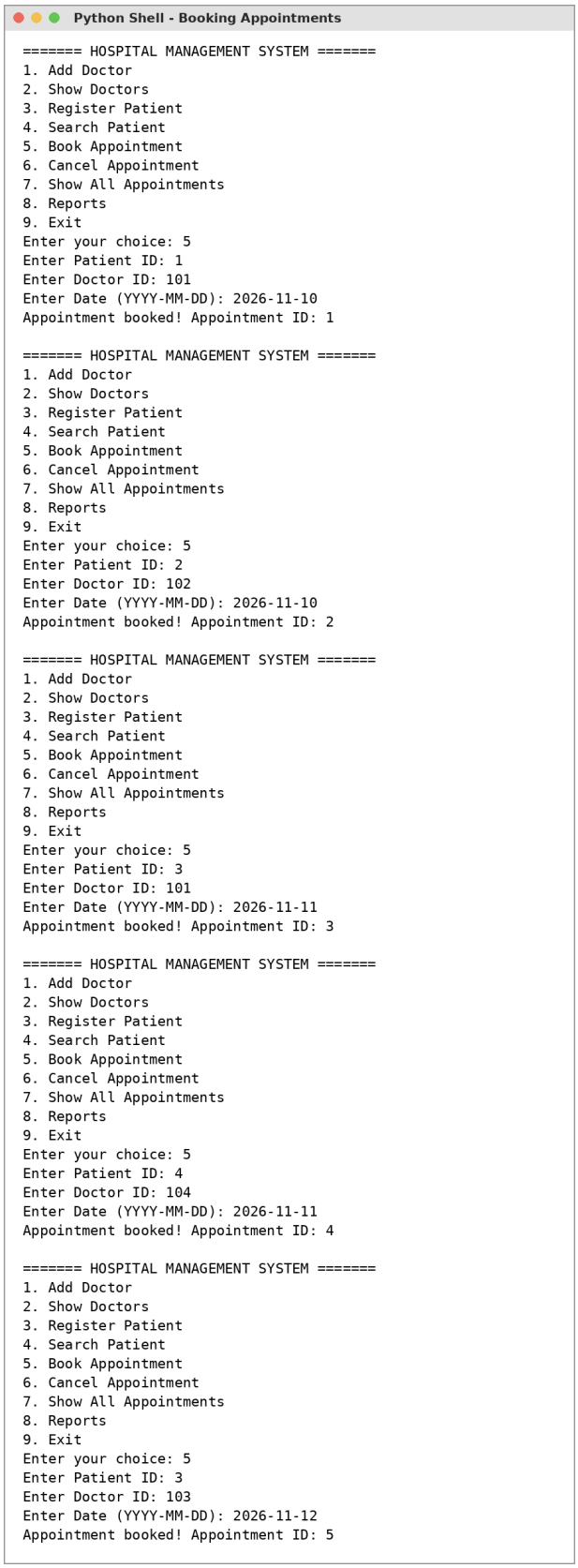
Python Shell - More Patients and Duplicate ID Error

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 3
Enter Patient ID: 3
Enter Patient Name: Mohan Rawat
Enter Age: 67
Enter Gender (M/F): M
Enter Phone Number: 9758123456
Patient registered successfully!

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 3
Enter Patient ID: 4
Enter Patient Name: Sunita Joshi
Enter Age: 34
Enter Gender (M/F): F
Enter Phone Number: 9897011223
Patient registered successfully!

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 3
Enter Patient ID: 1
Enter Patient Name: Test
Enter Age: 20
Enter Gender (M/F): M
Enter Phone Number: 9000000000
Database ERROR: Duplicate entry '1' for key 'PRIMARY'
```

Output 4: Booking Appointments



Output 5: Cancel Appointment and Show All (JOIN)

Python Shell - Cancel Appointment and Show All (JOIN)

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 6
Enter Appointment ID to cancel: 4
Appointment 4 cancelled.

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 7
ID   Patient          Doctor             Department   Date        Status
-------------------------------------------------------------------------
2    Kavya Bisht      Dr. Meena Joshi    Pediatrics   2026-11-10  Booked
1    Ramesh Negi      Dr. R.K. Sharma    Cardiology   2026-11-10  Booked
4    Sunita Joshi     Dr. Sunita Bisht   General      2026-11-11  Cancelled
3    Mohan Rawat      Dr. R.K. Sharma    Cardiology   2026-11-11  Booked
5    Mohan Rawat      Dr. Anil Rawat     Orthopedics  2026-11-12  Booked
```

Output 6: Search Patient (LIKE)

Python Shell - Search Patient (LIKE)

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 4
Enter name (or part of name) to search: joshi
ID     Name                  Age Gender Phone
-------------------------------------------------
4      Sunita Joshi           34      F 9897011223

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 4
Enter name (or part of name) to search: ra
ID     Name                  Age Gender Phone
-------------------------------------------------
1      Ramesh Negi            45      M 9876543210
3      Mohan Rawat            67      M 9758123456
```

Output 7: Reports (GROUP BY and Aggregate Functions)

Python Shell - Reports (GROUP BY and Aggregate Functions)

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 8

--- Appointments per Department (GROUP BY) ---
Cardiology     : 2
Orthopedics    : 1
Pediatrics     : 1

--- Fees Collected by each Doctor (SUM) ---
Dr. R.K. Sharma      : Rs. 1600
Dr. Anil Rawat       : Rs. 600
Dr. Meena Joshi      : Rs. 500

--- Patient Statistics (MIN, MAX, AVG) ---
Total Patients: 4
Youngest: 8 | Oldest: 67 | Average Age: 38.5
```

Output 8: Invalid Input and Exit

Python Shell - Invalid Input and Exit

```
======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 5
Enter Patient ID: abc
ERROR: Please enter numbers where required.

======= HOSPITAL MANAGEMENT SYSTEM =======
1. Add Doctor
2. Show Doctors
3. Register Patient
4. Search Patient
5. Book Appointment
6. Cancel Appointment
7. Show All Appointments
8. Reports
9. Exit
Enter your choice: 9
Database connection closed. Goodbye!
```

Technological Requirements: Hardware & Software

Software

- Python 3.x (IDLE)
- MySQL Server 8.0
- mysql-connector-python module
- VS Code / any text editor

Hardware

- Laptop / PC with Intel Core i3 or above
- 4 GB RAM or more
- Keyboard and Mouse

References

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